EAST WEST UNIVERSITY

Project Report

A Multi-Objective Neural Architecture Search Framework Using Genetic Algorithms for Survival Prediction in Lung Cancer

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1. Introduction

The use of machine learning for clinical outcome prediction has seen rapid growth in recent years, especially in oncology, where timely and accurate prognosis can significantly influence treatment planning. Lung cancer continues to be one of the leading causes of cancer-related deaths globally. Predicting patient survival using structured clinical data represents both a critical clinical need and a compelling machine learning problem. Although manually designed neural networks have shown promising results in such tasks, their development often demands considerable domain expertise and repeated trial-and-error experimentation, which introduces subjectivity and reduces efficiency.

Neural Architecture Search (NAS) provides a more systematic and automated approach by optimizing neural network architectures within a defined search space. Instead of depending on human judgment to determine hyperparameters such as the number of layers or neurons per layer, NAS allows an optimization algorithm to explore these decisions. Among various NAS techniques—including reinforcement learning, Bayesian optimization, and evolutionary algorithms—Genetic Algorithms (GA) stand out for their interpretability, robustness in handling complex, non-convex search spaces, and their ability to operate without gradient information. Drawing inspiration from natural selection, Genetic Algorithms evolve a population of candidate architectures through processes of selection, crossover, and mutation, gradually improving performance over successive generations.

In this study, we propose a Genetic Algorithm-based Neural Architecture Search framework for predicting survival outcomes in lung cancer patients using structured clinical data. The search space focuses on two essential architectural parameters: the number of hidden layers and the number of neurons in each layer, both of which significantly affect model capacity and generalization ability. Candidate architectures are evaluated primarily based on classification performance using the Area Under the ROC Curve (AUC), with accuracy, loss, and training time serving as secondary metrics.

This work makes three main contributions. First, we develop a comprehensive GA-based NAS pipeline specifically designed for binary classification on tabular healthcare data. Second, we systematically compare the performance of NAS-discovered architectures against a manually designed baseline model across multiple evaluation criteria, including accuracy, loss, AUC, and computational efficiency. Third, we examine the evolutionary behavior of the genetic algorithm to understand its convergence patterns across generations. The findings suggest that automated architecture search can be a practical and effective substitute for manual model design, even with limited computational resources.

2. Methodology

2.1 Dataset and Preprocessing

The dataset employed in this study comprises structured clinical records of lung cancer patients, with the binary target variable “survived” indicating whether the patient survived after treatment. It includes a variety of demographic, clinical, and lifestyle features such as age, gender, country, cancer stage, body mass index (BMI), cholesterol levels, smoking status, family history of cancer, comorbidities (e.g., hypertension, asthma, cirrhosis), and type of treatment received.

Several preprocessing steps were performed to ensure data quality and suitability for modeling. Irrelevant columns, including unique identifiers (id) and date-related fields (diagnosis_date and end_treatment_date), were removed to avoid data leakage and reduce dimensionality. Missing values in numerical features were imputed using the median, while categorical features were imputed with the most frequent category. All categorical variables were then transformed using one-hot encoding with the drop_first=True parameter to prevent multicollinearity.

The dataset showed notable class imbalance, with approximately 78% non-survivors and 22% survivors. To mitigate this issue, the minority class was upsampled using random resampling with replacement, resulting in a balanced dataset. Class weighting was also investigated in supplementary experiments. All features were standardized using StandardScaler to normalize the input distributions and enhance training stability of the neural networks. All input features were standardized using StandardScaler. This step normalizes the data distribution, improving convergence during training.

For computational efficiency during the NAS process, a subset of 10,000 samples was utilized for architecture search, while the complete dataset was used for final baseline evaluation. The data was divided into training and testing sets using an 80:20 stratified split to maintain the original class distribution.

2.2 Search Space Definition

The search space in this study is defined based on two key architectural components: the number of hidden layers and the number of neurons in each layer. Specifically, the number of hidden layers (L) is selected from the set ({1, 2, 3}), allowing the exploration of shallow to moderately deep neural networks. For each layer, the number of neurons (N_i) is independently chosen from the set ({16, 32, 64, 128}). Unlike uniform-width architectures, this design permits different numbers of neurons across layers, enabling more flexible and expressive network structures. Each candidate architecture is encoded as a fixed-length vector ((L, N_1, N_2, N_3)), where only the first (L) layers are active and the remaining positions are ignored. This results in a total of (3 \times 4^3 = 192) possible architecture combinations, providing a compact yet sufficiently diverse search space for optimization. All generated architectures utilize the ReLU activation function in hidden layers and a sigmoid activation function in the output layer. The models are trained using the Adam optimizer with binary cross-entropy loss, which is well-suited for the binary classification task of lung cancer survival prediction.

2.3 Genetic Algorithm Framework

A Genetic Algorithm (GA) is used to efficiently explore the defined search space.

Representation: Each individual in the population is encoded as a tuple (L,N1,N2,N3), representing the depth of the network and the number of neurons in each layer.

Initialization: An initial population of 10 individuals is randomly generated by sampling from the search space.

Fitness Evaluation: Each candidate architecture is instantiated as a neural network and trained for a limited number of epochs (10 epochs) to keep the search computationally tractable. Performance is primarily assessed using accuracy on the test set. Loss and training time are recorded as additional metrics. A composite fitness function is used to balance predictive performance and computational cost.

Selection: Truncation selection is applied, retaining the top 50% performing individuals for reproduction. Elitism is implemented by directly carrying the best-performing individual to the next generation.

Crossover: Uniform crossover is performed, where each parameter (N1,N2,N3) is independently inherited from one of the two parents with equal probability.

Mutation: With a mutation probability of 0.2, a randomly selected layer's neuron value is replaced by another value from the search space. This helps maintain population diversity and prevents premature convergence.

Termination: The algorithm is run for 5 generations. The best-performing architecture observed across all generations is selected as the final output of the NAS process.

To manage computational constraints, reduced training epochs and a sampled dataset subset are used during the search, accepting a trade-off between search speed and full model convergence.

2.4 Baseline Model

A manually designed baseline neural network is implemented for comparison. The baseline consists of two hidden layers with 64 and 32 neurons respectively, followed by a sigmoid output layer. This configuration represents a typical architecture commonly used for tabular classification problems.

The baseline model is trained under the same conditions (optimizer, loss function, batch size, and number of epochs) as the NAS-discovered models to ensure a fair and consistent comparison.

2.5 Evaluation Metrics

Model performance is assessed using the following metrics:

- **Accuracy**: Overall proportion of correct predictions
- **Binary Cross-Entropy Loss**: Quantifies the confidence and calibration of predictions
- **Training Time**: Wall-clock time in seconds, measured during model training
- **AUC-ROC**: The primary performance metric, particularly suitable for imbalanced classification tasks

In addition, confusion matrices and ROC curves are generated to provide visual insights into classification behavior.

3. Experimental Results and Analysis

3.1 Overview of Experimental Setup

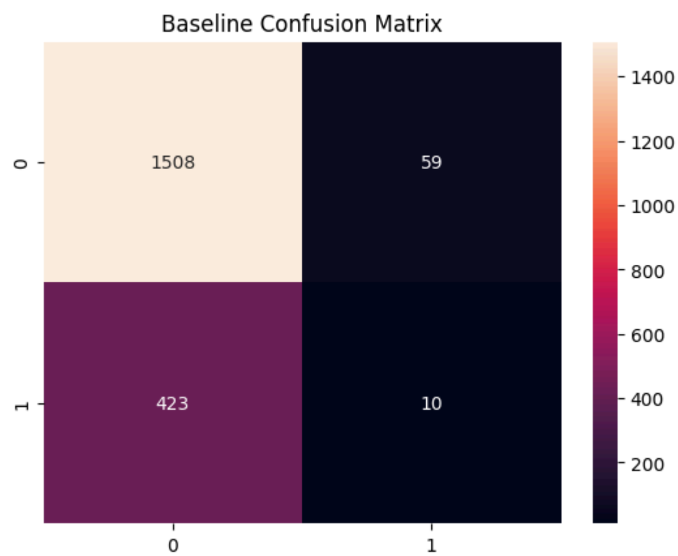
The experimental evaluation was conducted to assess the effectiveness of the proposed Genetic Algorithm-based Neural Architecture Search (NAS) framework in predicting lung cancer survival outcomes. All models were implemented using TensorFlow and Keras to ensure consistency in training and evaluation.

To maintain fairness, both the baseline model and NAS-generated architectures were trained under identical conditions, including the same optimizer (Adam), loss function (binary cross-entropy), and training configuration. During the NAS process, models were trained for a limited number of epochs to reduce computational cost, while final evaluations were conducted more thoroughly.

3.2 Baseline Model Performance

The manually designed baseline neural network was used as a reference for comparison. This model achieved an accuracy of approximately 75.9% on the test dataset. The baseline architecture consists of 2 hidden layers, 64 and 32 neurons respectively (total 96 neurons).

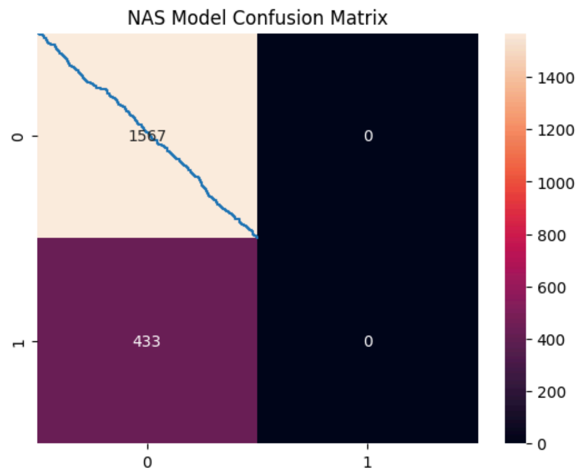
However, a deeper analysis using AUC-ROC reveals that the model's discriminative ability is limited, with an AUC score of approximately 0.53. This indicates that the model performs only slightly better than random guessing in distinguishing between survivors and non-survivors.



3.3 Performance of NAS-Generated Architectures

The Genetic Algorithm-based NAS framework explored multiple neural network architectures across several generations. The best-performing architecture achieved an accuracy of approximately 78.3%, representing a slight improvement over the baseline model.

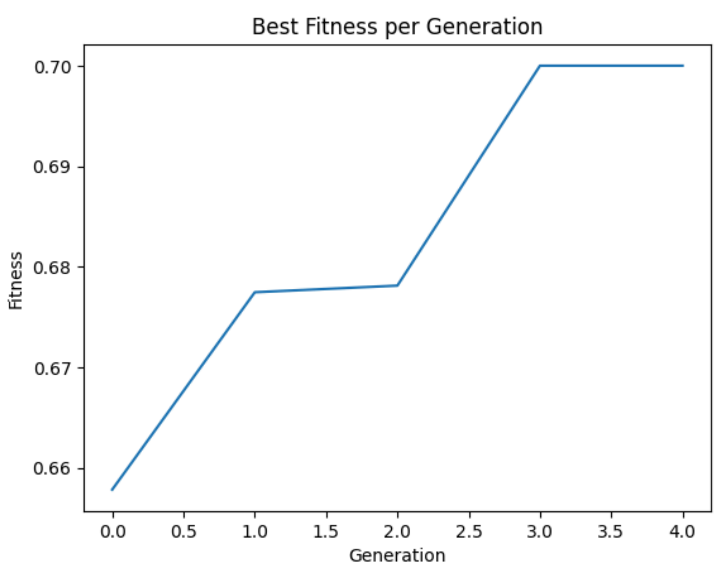
In terms of AUC-ROC, the NAS-generated models achieved values in the range of 0.54 to 0.56, indicating a marginal improvement in classification capability.



3.4 Evolutionary Behavior of the Genetic Algorithm

The genetic algorithm demonstrated stable convergence behavior across generations. The fitness value showed gradual improvement and stabilized after a few generations, indicating convergence within the limited search space.

However, the variation in performance among different architectures remained small, suggesting limited diversity within the search space and a tendency of the algorithm to converge toward similar solutions.



Dataset Limitations and Impact on NAS Performance

An important factor influencing the performance of the proposed NAS framework is the nature of the dataset itself. The dataset exhibits characteristics that make it less suitable for effective optimization using Genetic Algorithms. In particular, the presence of class imbalance leads the model to favor the majority class, resulting in high accuracy but poor discriminative performance, as reflected by low AUC scores.

Additionally, the dataset appears to have limited feature separability, meaning that the input features do not provide strong distinguishing patterns between the classes. This restricts the ability of different neural architectures to learn meaningful decision boundaries, causing many candidate models to produce similar performance regardless of their structural differences.

Furthermore, the relatively low sensitivity of model performance to architectural variations reduces the effectiveness of the search process. Since different architectures yield only marginal performance differences, the Genetic Algorithm has limited guidance for selecting significantly better solutions, leading to early convergence.

These factors collectively suggest that the dataset plays a critical role in constraining the performance of the NAS framework. Addressing these issues through techniques such as data balancing, feature engineering, and improved data quality could significantly enhance the effectiveness of the proposed approach.

3.5 Comparative Analysis with Alternative Models

To provide additional context, a traditional machine learning model using XGBoost was also evaluated. This model achieved an accuracy of approximately 74.9%, which is lower than both the baseline and NAS-based neural network models.

- XGBoost Accuracy: ~74.9%
- Neural networks perform slightly better
- Difference is not significantly large

3.6 Discussion of Results

The experimental results demonstrate that the proposed Genetic Algorithm-based Neural Architecture Search (NAS) framework is capable of automatically exploring and identifying neural network architectures within a predefined search space. While the improvement in accuracy over the baseline model is relatively small, the framework successfully illustrates the potential of evolutionary optimization for automated model design.

A key observation from the results is the discrepancy between accuracy and AUC scores. Although the models achieve moderately high accuracy (around 78%), the relatively low AUC values indicate limited discriminative capability. This suggests that the models tend to favor the majority class, which is a common challenge in imbalanced classification problems. As a result, accuracy alone does not fully reflect the true performance of the model.

The modest performance improvement can be attributed to several factors. First, the search space is relatively constrained, limiting the diversity of architectures explored by the algorithm. Second, the use of a reduced number of training epochs during the search phase restricts the ability of candidate models to fully converge. Third, inherent characteristics of the dataset, including class imbalance and feature limitations, further impact model performance.

Despite these limitations, the study provides valuable insights into the behavior of NAS in practical scenarios. It highlights the importance of carefully designing the fitness function, incorporating appropriate evaluation metrics, and ensuring sufficient exploration within the search space to achieve better results.

4. Conclusion

This study presented a Genetic Algorithm-based Neural Architecture Search (NAS) framework for lung cancer survival prediction using structured clinical data. The proposed approach aims to automate the process of neural network design, reducing the need for manual architecture tuning.

The experimental findings indicate that the NAS framework is capable of identifying architectures that achieve performance comparable to, and slightly better than, a manually designed baseline model. Although the overall improvement is modest and the AUC-ROC scores indicate limited discriminative power, the results demonstrate the feasibility of applying evolutionary NAS techniques in healthcare prediction tasks.

Importantly, this work emphasizes that model performance is influenced not only by architecture design but also by dataset characteristics, class imbalance, and evaluation strategies. The study therefore serves as a foundational step toward more advanced NAS implementations in medical applications.

For future work, several enhancements can be considered, including expanding the search space, increasing the number of generations, incorporating multi-objective optimization strategies, and applying data balancing and feature engineering techniques. These improvements have the potential to significantly enhance the effectiveness of the NAS framework and lead to more robust predictive models.